Problem

Problem · The check digit of an ISBN-13 code. An ISBN-13 code is a sequence of thirteen digits $a_1 a_2 \ldots a_{13}$. Let

$$S = a_1 + 3a_2 + a_3 + 3a_4 + \cdots + 3a_{12} + a_{13},$$

with the weights 1 and 3 alternating. The code is valid if and only if $S \equiv 0 \pmod{10}$; the last digit a_{13}, called the check digit, is chosen to make it so.

1. Consider the fictitious number 978-2-0601-2345-a_{13}. Compute the check digit a_{13}.
2. Assume that a single digit a is replaced by a digit $a' \neq a$.
   (a) Prove that S changes by $w(a' - a)$, with $w \in \{1, 3\}$.
   (b) Check that $7 \times 3 \equiv 1 \pmod{10}$ and deduce that $3(a' - a) \equiv 0 \pmod{10}$ implies $a' - a \equiv 0 \pmod{10}$.
   (c) Conclude: an error in a single digit is always detected.
3. Assume that two distinct adjacent digits a and b are swapped.
   (a) Prove that S changes by $2(b - a)$ or by $2(a - b)$.
   (b) Determine under which condition on a and b the error is not detected. Give an example.